

Conclusions on Advanced Database Systems

We live in a data-driven world

- Data = Currency, Data = Power, Data = Fun 😊
- Data is supposed to change the mode of operation for almost every single field
- Science, Technology, Healthcare, Genomics, Business, Manufacturing, Journalism, Government, Sports, Education, ...
- We must know how to collect, store, process, and analyze such data, **big and small**

What we have seen

- Structured (relational) data
- Big and small data alike
- SQL
- OLAP & OLTP

Journey from traditional (disk-based, centralized) Database Management Systems (the norm for decades, still major role) to parallel DBMS and other modern database architectures (NoSQL).

We relayed some key messages

- Importance of query optimization: most of today's popular DBMSs, PDBMs, NoSQL systems need a query optimizer
- Importance of declarative languages (SQL)

We charted the limitations of RDBMS

Relational databases have been around for decades, and they are very well designed for...

- ... structured data
- ... often/concurrent read/writes, integrity (OLTP)

Their core concepts were designed also decades ago

- Data / schema normalization :
 - Removing redundancy / duplication and optimizing storage
- Originally designed for single machines
 - Scaling usually mean to buy a new and “bigger” machine

Lots of things changed since RDBMS inception

- Dropping cost of disks, SSDs, RAM
 - Cheaper to store everything than to figure out what to throw away !
- Nature of data being collected
 - From data that's *obviously* valuable to data whose value is less apparent
- Rise of social media and user-generated content
 - Large increase in data volume and user base, need for data analytics
- Growing maturity of data mining techniques
 - Demonstrates value of data analytics

A glimpse on systems

Many kinds of systems optimized for various kinds of workloads and manipulations



Comparing data management systems

When you study a new system, compare it with RDBMS on

- data model
- consistency mechanisms
- storage mechanisms, replication
- durability guarantees
- concurrency control, transactions
- availability
- query support
- performance (latency, throughput, workload management, scalability)

One size does not fit all...

- Challenges around managing in a DBMS large or increasing data volumes (e.g., analytical) are different from the ones around maintaining response times (latency) as the number of users (concurrent transactions) increases
- Many different kinds of boats to sail the data deluge !



Remaining milestones

- Practical Assignment 2 – to be finalized and submitted this week (Nov 16)
- Practical Assignment 3 – Concurrency control with PostgreSQL and Oracle (Nov 23)
- Practical Assignment 4 – NoSQL with Apache Cassandra (Dec 7)
- Final exam – December 4

Capstone projects in databases

- Distributed and Scalable Data Management with Spark and Cassandra
- Composable Graph Database System Design for Information Diffusion Graphs.

Thank you all !